```
create database task1;
use task1;

create table departments (
   dept_id int,
   dept_name varchar(50)
);

insert into departments values
(1, 'IT'),
(2, 'HR'),
(3, 'Finance'),
(4, 'Marketing'),
(5, 'Sales'),
(6, 'Operations');

create table employees (
   emp_id int,
   emp_name varchar(50),
   dept_id int,
   salary int,
   manager_id int,
   join_date date
);

insert into employees values
(101, 'Rahul', 1, 90000, null, '2021-01-10'),
(102, 'Priya', 1, 70000, 101, '2022-03-15'),
(103, 'Arjun', 1, 70000, 101, '2022-06-20'),
(104, 'Sneha', 1, 60000, 101, '2023-02-10'),

(105, 'Kiran', 2, 65000, null, '2021-07-12'),
(106, 'Anjali', 2, 50000, 105, '2023-01-05'),
(107, 'Ravi', 2, 45000, 105, '2023-04-18'),

(108, 'Vikram', 3, 80000, null, '2020-09-18'),
(109, 'Neha', 3, 60000, 108, '2022-11-25'),
(110, 'Aman', 3, 60000, 108, '2023-05-10'),

(111, 'Meena', 4, 55000, null, '2022-02-14'),
(112, 'Varun', 4, 40000, 111, '2023-08-20'),

(113, 'Suresh', 5, 75000, null, '2021-05-30'),
(114, 'Divya', 5, 50000, 113, '2022-09-15'),
(115, 'Rohan', 5, 50000, 113, '2023-03-12');

create table projects (
   project_id int,
   project_name varchar(50),
   dept_id int,
   budget int
);

insert into projects values
(201, 'Website', 1, 500000),
(202, 'Mobile App', 1, 700000),
(203, 'Recruitment', 2, 200000),
```

```
(204, 'Audit System', 3, 400000),
(205, 'Sales Portal', 5, 600000);


create table employeeprojects (
   emp_id int,
   project_id int,
   hours_worked int
);

insert into employeeprojects values
(101, 201, 120),
(102, 201, 100),
(103, 201, 100),
(104, 201, 80),

(101, 202, 150),
(102, 202, 120),
(103, 202, 120),

(105, 203, 100),
(106, 203, 90),
(107, 203, 90),

(108, 204, 140),
(109, 204, 110),
(110, 204, 110),

(113, 205, 160),
(114, 205, 120),
(115, 205, 120);

create table orders (
   order_id int,
   customer_id int,
   order_date date,
   amount decimal(10,2),
   status varchar(20)
);

insert into orders values
(1, 501, '2024-01-10', 30000, 'Completed'),
(2, 501, '2024-02-15', 40000, 'Completed'),
(3, 501, '2024-03-20', 35000, 'Completed'),

(4, 502, '2024-01-12', 20000, 'Completed'),
(5, 502, '2024-02-18', 15000, 'Completed'),

(6, 503, '2024-01-05', 10000, 'Completed'),
(7, 503, '2024-03-10', 15000, 'Completed'),

(8, 504, '2024-04-15', 60000, 'Completed'),

(9, 505, '2024-05-10', 120000, 'Completed'),

(10, 506, '2024-01-20', 5000, 'Pending'),
(11, 506, '2024-02-20', 7000, 'Completed'),
```

```
(12, 507, '2024-06-10', 25000, 'Completed'),
(13, 507, '2024-07-15', 30000, 'Completed'),
(14, 507, '2024-08-20', 10000, 'Completed'),

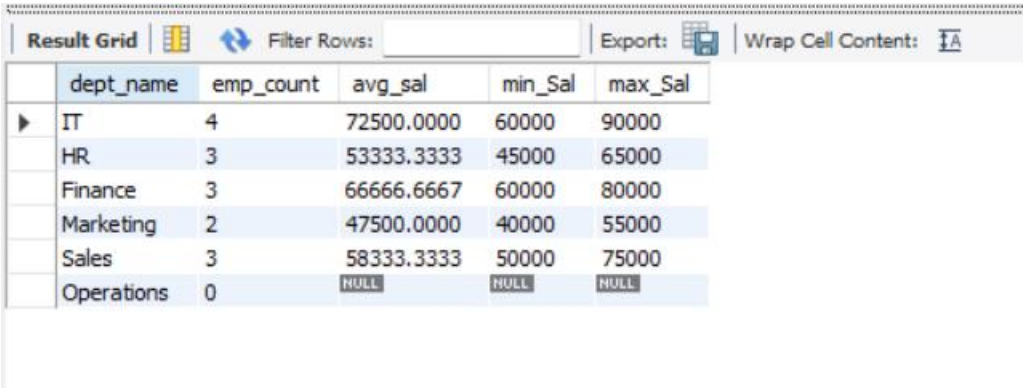
(15, 508, '2024-02-10', 45000, 'Completed'),
(16, 508, '2024-03-12', 10000, 'Completed'),
(17, 508, '2024-04-18', 5000, 'Completed'),

(18, 509, '2024-01-15', 100000, 'Completed'),
(19, 509, '2024-02-15', 50000, 'Completed'),

(20, 510, '2024-03-10', 15000, 'Cancelled');


-- select * from employees;
-- select * from departments;
-- select * from employeeprojects;
-- select * from orders;

-- 1.  Department Salary Summary
select d.dept_name,count(e.emp_id) as emp_count,avg(e.salary) as avg_sal,min(e.salary) as
min_Sal,max(e.salary) as max_Sal from departments d
left join employees e on d.dept_id=e.dept_id group by d.dept_id,d.dept_name;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	emp_count	avg_sal	min_Sal	max_Sal
IT	4	72500.0000	60000	90000
HR	3	53333.3333	45000	65000
Finance	3	66666.6667	60000	80000
Marketing	2	47500.0000	40000	55000
Sales	3	58333.3333	50000	75000
Operations	0	NULL	NULL	NULL

```
-- 2. Second Highest Salary per Department
select emp_name, dept_id, salary from (select emp_name,dept_id,salary,dense_rank() over(partition
by dept_id order by salary desc) as rnk from employees) x where rnk = 2;
```



Result Grid | Filter Rows:

emp_name	dept_id	salary
Priya	1	70000
Arjun	1	70000
Anjali	2	50000
Neha	3	60000
Aman	3	60000
Varun	4	40000
Divya	5	50000
Rohan	5	50000

-- 3. Employees Above Department Average
select emp_name, dept_id, salary from employees e where salary > (select avg(salary)from employees where dept_id = e.dept_id);

Result Grid | Filter Rows:

emp_name	dept_id	salary
Rahul	1	90000
Kiran	2	65000
Vikram	3	80000
Meena	4	55000
Suresh	5	75000

-- 4. Manager vs Team Salary
select m.emp_name as manager_name,count(e.emp_id) as direct_reports,sum(e.salary) as total_team_salary,m.salary as manager_salary from employees m
join employees e on m.emp_id = e.manager_id group by m.emp_id, m.emp_name, m.salary having count(e.emp_id) >= 2;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

manager_name	direct_reports	total_team_salary	manager_salary
Rahul	3	200000	90000
Kiran	2	95000	65000
Vikram	2	120000	80000
Suresh	2	100000	75000

-- 5. Top Project Contributors
select project_id, emp_id, total_hours from (
  select project_id, emp_id,sum(hours_worked) as total_hours,dense_rank() over (partition by project_id order by sum(hours_worked) desc) as rnk
  from employeeprojects group by project_id, emp_id) x
where rnk <= 2;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

project_id	emp_id	total_hours
201	101	120
201	102	100
201	103	100
202	101	150
202	102	120
202	103	120
203	105	100
203	106	90
203	107	90
204	108	140

-- 6.Departments Without Active Projects

```
select d.dept_id, d.dept_name from departments d join employees e on d.dept_id = e.dept_id left
join projects p on d.dept_id = p.dept_id
where p.project_id is null group by d.dept_id, d.dept_name;
```

Result Grid | Filter Rows:

dept_id	dept_name
4	Marketing

-- 7. Running Order Revenue

```
select order_date,sum(amount) as daily_revenue,sum(sum(amount)) over (order by order_date) as
cumulative_revenue
from orders where status = 'Completed' group by order_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_date	daily_revenue	cumulative_revenue
2024-01-05	10000.00	10000.00
2024-01-10	30000.00	40000.00
2024-01-12	20000.00	60000.00
2024-01-15	100000.00	160000.00
2024-02-10	45000.00	205000.00
2024-02-15	90000.00	295000.00
2024-02-18	15000.00	310000.00
2024-02-20	7000.00	317000.00
2024-03-10	15000.00	332000.00
2024-03-12	10000.00	342000.00

Result 7

-- 8. Customer Order Classification

```
select customer_id,sum(amount) as total_amount,
   case
      when sum(amount) >= 100000 then 'PLATINUM'
      when sum(amount) >= 50000 then 'GOLD'
      when sum(amount) >= 20000 then 'SILVER'
      else 'BRONZE'
   end as customer_class
from orders where status = 'Completed' group by customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

customer_id	total_amount	customer_class
501	105000.00	PLATINUM
502	35000.00	SILVER
503	25000.00	SILVER
504	60000.00	GOLD
505	120000.00	PLATINUM
506	7000.00	BRONZE
507	65000.00	GOLD
508	60000.00	GOLD
509	150000.00	PLATINUM

-- 10.Salary Gap Analysis
select dept_id,emp_name,salary,
max(salary) over(partition by dept_id) - min(salary) over(partition by dept_id) as salary_gap from employees;


Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_id	emp_name	salary	salary_gap
1	Rahul	90000	30000
1	Priya	70000	30000
1	Arjun	70000	30000
1	Sneha	60000	30000
2	Kiran	65000	20000
2	Anjali	50000	20000
2	Ravi	45000	20000
3	Vikram	80000	20000
3	Neha	60000	20000
3	Aman	60000	20000